Le Nghia Hiep

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Summary

A Computer Engineering student with a sharp systems-oriented mindset and a passion for core technology research. I possess a strong technical foundation, which enables me to quickly adapt to and master new technologies to solve complex problems. Beyond technical expertise, I take pride in my strong communication, presentation, and teamwork skills.

Education

Hanoi University of Science and Technology (HUST) — Hanoi, Vietnam
Majoring in Computer Engineering - CPA 3.60 — *Sep. 2023 – Present*

Phu Xuyen A High School — Phu Xuyen, Hanoi, Vietnam
Achieved 28.7 scores (Group A00) in the High School Graduation Examination — *2020 – 2023*

Experience

MDC Software
Embedded & Firmware Engineer — *July, 2026 – Present*

- Hardware Design & Prototyping: Designed electrical schematics, custom 3D-printed mechanical enclosures, power management circuitry, and performed board soldering/assembly for multi-module integration.
- Firmware Architecture (ESP-IDF): Developed core firmware on ESP32-S3 (N16R8), handling real-time task scheduling, state machines, and low-latency cloud communication workflows for AI voice interaction.
- Audio & Multimedia Pipeline: Integrated full-duplex audio using INMP441 (I2S Mic) and MAX98357A (I2S DAC/Amp) with 3W speaker, along with visual feedback driving a ST7789 IPS LCD and OV3660 camera.
- Actuation & Sensor Integration: Configured dual DRV8833 H-bridge motor drivers for N20 geared motors, integrated VL53L0X ToF distance sensors for obstacle detection, and handled capacitive touch (TTP223) / UI inputs.
- Connectivity & Maintenance: Implemented Wi-Fi, Bluetooth, USB communication, and OTA firmware updates to ensure seamless remote configuration, testing, and deployment.

Modern Computer Networks and Communications Technology Lab - NetLab — SoICT - HUST
Research Assistant — *April, 2025 – July, 2026*

- Research on AIoT in Healthcare
- Biosignal Research (EEG, ECG)

Technical Skills

Language: C/C++ (Embedded System), Python (Data processing & AI), Java, JS/HTML/CSS
Embedded Systems & Hardware: ESP32 development, Sensors integration
Networking & Protocols: IoT Protocols (MQTT, HTTP/Restful API), Network Basics (TCP/IP, UDP)
Tools & Platforms: Git/Github, Arduino IDE/PlatformIO
English: Intermediate (TOEIC Equivalent 650-700, enough to read and understand AI papers and documentation)

Projects

SmartClock - AI-Powered IoT Health & Environment Monitor
ESP32, PlatformIO, FreeRTOS, Python, FastAPI, MQTT, Scikit-Learn, SQLite, Kotlin — Full-stack AIoT project

- Designed and architected an end-to-end IoT ecosystem connecting edge hardware, an AI-powered cloud backend, and a native mobile application.
- Developed non-blocking firmware for the ESP32-C3 microcontroller using asynchronous timing loops, ensuring high-frequency sensor polling (MAX30102, SHT31) and smooth 33 FPS OLED rendering without thread-blocking.
- Implemented a hybrid communication pipeline integrating real-time time-series data streaming via MQTT with secure user profile synchronization via RESTful APIs (FastAPI).
- Trained and deployed a Machine Learning model (Scikit-Learn) on a local Python server to process raw infrared (IR) PPG signals and return real-time health and recovery assessments.

- Engineered a responsive Android application using Kotlin and Jetpack Compose under the MVVM pattern, incorporating live data binding and dynamic canvas-based PPG waveform animations synchronized with real-time BPM.
- Optimized hardware power consumption by programming ESP32 deep-sleep modes, utilizing a capacitive touch button as an external GPIO hardware wake-up source.
- Device images: *Direct link to circuit image*
- Github: *link*

AI-Driven ECG Arrhythmia Classifier

ESP32, MQTT, Deep Learning, TensorFlow, Scikit-learn Full-stack AIoT project

- Hardware Design & Integration: Designed schematics and developed prototypes integrating ESP32 microcontrollers, AD8232, and MPU-6050 sensors; optimized power management systems using TP4056 charging circuits and LiPo batteries.
- Communication Protocols & Connectivity: Implemented real-time data streaming using the MQTT protocol over wireless networks to ensure stable, low-latency communication between devices and the cloud.
- Hybrid Deep Learning Architecture: Designed and implemented a hybrid 1D CNN-LSTM model to classify cardiac arrhythmias. Used 1D CNN for spatial feature extraction from raw ECG signals and LSTM to capture long-term temporal dependencies in heart rate variability
- Signal Processing & Optimization: Developed a preprocessing pipeline to handle raw signal noise (Baseline Wander, Powerline Interference) using digital filters. Conducted data segmentation and normalization on the MIT-BIH Arrhythmia Database to ensure high model generalization
- Device images: *Direct link to circuit image*
- Source code: *CNN-1D-and-LSTM-for-Arrythmia-Classification*

RAG-based University Query Chatbot

RAG, Python, LangChain, LLMs, FAISS, Streamlit RAG Chatbot

- Data Pipeline Development: Engineered a robust pipeline to extract and preprocess data (including advanced chunking) from complex academic regulation PDFs
- Retrieval Optimization: Leveraged Recursive Character Text Splitting and Embedding Models to transform unstructured text into high-dimensional vectors, managed within a FAISS vector database
- Advanced RAG Implementation: Integrated Context Retrieval to provide automated source citations for every response, ensuring transparency and factual accuracy for legal and regulatory information
- Deployment: Developed and launched an interactive user interface using Streamlit to facilitate seamless student-bot interaction
- Source code: *AI-chatbot-for-HUST-students*

Honors And Awards

- *Top 10 — Kawai Business Startup 2026*
- *Second Prize in the Student Creative Ideas Challenge - SoICT 2025*
- *Awarded Academic Achievement scholarship of SoICT*
- *Awarded Lotte Scholarship, SCG Sharing the Dream Scholarship 2023, 2024*